

Raphaël Terrine

3rd year PhD student at Polytechnique Institute

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Research Interests

I am interested in the study of Partial Differential Equations both numerically and theoretically and the study of inverse problems using optimal control and one shot methods. I'm also interested in the implementation and the analysis of the Finite Elements Method.

Education

2023 – 2025 PhD in Applied Mathematics ENSTA Paris/Polytechnique

Thesis: Identification of bottom deformations of the ocean from surface measurements.

2020 – 2023 Engineering school ENSTA Paris

Specialisation : Applied maths and waves propagation

2022 – 2023 Master's degree in applied Mathematics Polytechnique Institute

Specialisation : Modelisation, Simulation and Analysis

2018 – 2020 French Preparatory Classes MPSI-MP* Lycée Henri IV

Two-year intensive preparatory program for competitive entrance to French Grandes Écoles

Professional experiences

Summer 2023 Pre PhD internship at UMA ENSTA Paris

Identification of bottom deformations of the ocean from surface measurements

Summer 2022 Research Internship supervised by G. Bal University of Chicago

Analysis of transport along the boundaries of domains generated by Coriolis forces in geophysical fluid flows.

Publications

2024 L. Bourgeois, J-F. Mercier, R. Terrine Inverse Problems and Imaging

Identification of bottom deformations of the ocean from surface measurements.

In preparation L. Bourgeois, P. Moireau, R. Terrine

About convergence of a discretized optimal control approach: application to a tsunami inverse problem.

Talks

29 October 2025 Invited speaker at PICOF'2025

Hamammet

Convergence of a numerical scheme for solving an inverse problem : Application to tsunami reconstruction.

5 June 2025 SMAI 2025

Carcans

Identification of bottom deformations from surface measurements

4 July 2024 Waves 2024

Berlin

About the identification of tsunamis from surface measurements,

Teaching

2023-2025 Introduction to tools for PDE

ENSTA Paris

L3 level class

Technical Skills

Programming: Python, C++, MATLAB

Tools: LaTeX, Git

Languages: French (native), English (fluent)